

Load Interconnection Request System Impact Study Report

For Queue Project Q20-1633

Tillery 138 kV
January 24, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1633. The Interconnection Customer proposes a 210.5 MW large load (data center) facility interconnecting at the Tillery 138 kV substation in the Pennsylvania area. The request was received on December 25, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 376 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,346,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: EVERETT / PITTSFORD 138kV
Contingency: WOODVILLE / U S ALLIANCE COOSA PINES 138kV
Loading: 171.7% (Rating: 153 MVA)
DFAX: 71.1%
Pre-project loading: 73.6%
2. Facility: TORAY PLASTIC AMERICA'S CHP PLANT / PITTSFORD 138kV
Contingency: WOODVILLE SUBSTATION / U S ALLIANCE COOSA PINES 138kV L
Loading: 158.8% (Rating: 214 MVA)
DFAX: 100.0%
Pre-project loading: 60.7%
3. Facility: DORMONT SUBSTATION / TILLERY 138kV
Contingency: PITTSFORD / TORAY PLASTIC AMERICA'S CHP PLANT 138kV Line
Loading: 155.7% (Rating: 181 MVA)
DFAX: 100.0%
Pre-project loading: 39.6%
4. Facility: BUFFALO SEWER AUTHORITY / TILLERY 138kV
Contingency: WOODVILLE / U S ALLIANCE COOS PINES 138kV
Loading: 145.3% (Rating: 198 MVA)
DFAX: 100.0%
Pre-project loading: 38.9%
5. Facility: DORMONT / CAUSEWAY 138kV L
Contingency: PITTSFORD / TORAY PLASTIC AMERICA'S CHP PLANT 138kV
Loading: 142.8% (Rating: 211 MVA)
DFAX: 100.0%
Pre-project loading: 43.1%

6. Facility: BUFFALO SEWER AUTHORITY / PUNTA DEL MAR 7302 138kV
Contingency: U S ALLIANCE COOSA PINES / WOODVILLE 138kV
Loading: 141.3% (Rating: 212 MVA)
DFAX: 100.0%
Pre-project loading: 42.2%
7. Facility: CAUSEWAY / U S ALLIANCE COOSA PINES 138kV
Contingency: TORAY PLASTIC AMERICA'S CHP PLANT / PITTSFORD 138kV
Loading: 138.8% (Rating: 238 MVA)
DFAX: 100.0%
Pre-project loading: 50.1%
8. Facility: U S ALLIANCE COOSA PINES / WOODVILLE 138kV
Contingency: TORAY PLASTIC AMERICA'S CHP PLANT / PITTSFORD 138kV line
Loading: 138.3% (Rating: 246 MVA)
DFAX: 100.0%
Pre-project loading: 52.8%
9. Facility: CORNERSVILLE / PUNTA DEL MAR 7302 138kV
Contingency: U S ALLIANCE COOSA PINES / WOODVILLE 138kV Line
Loading: 138.2% (Rating: 226 MVA)
DFAX: 100.0%
Pre-project loading: 45.0%
10. Facility: HALL / CORNERSVILLE 138kV
Contingency: WOODVILLE / U S ALLIANCE COSA PINES 138kV line
Loading: 137.8% (Rating: 237 MVA)
DFAX: 100.0%
Pre-project loading: 48.8%
11. Facility: HALL / TORAY PLASTIC AMERICA'S CHP PLANT 138kV
Contingency: U S ALLIANCE COOSA PINES / WOODVILLE SUBSTATION 138kV
Loading: 137.3% (Rating: 248 MVA)
DFAX: 100.0%
Pre-project loading: 52.4%
12. Facility: BOYNTONVILLE / WHILLOCK 138kV Line
Contingency: CADILLAC RENEWABLE ENERGY / LAS CRUCS 3602 138kV
Loading: 116.5% (Rating: 256 MVA)
DFAX: 12.6%
Pre-project loading: 106.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EVERETT / PITTSFORD 138kV	Rebuild line with higher-capacity conductor	\$1,432,000	100% (\$1,432,000)
PITTSFORD SUBSTATION / TORAY PLASTIC AMERICA'S CHP PLANT 138kV L	Rebuild line with higher-capacity conductor	\$1,345,000	100% (\$1,345,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DORMONT SUBSTATION / TILLERY 138kV	Rebuild line with higher-capacity conductor	\$1,310,000	100% (\$1,310,000)
BUFFALO SEWER AUTHORITY / TILLERY 138kV	Rebuild line with higher-capacity conductor	\$1,980,000	100% (\$1,980,000)
DORMONT / CAUSEWAY 138kV	Rebuild line with higher-capacity conductor	\$928,000	100% (\$928,000)
BUFFALO SEWER AUTHORITY / PUNTA DEL MAR 7302 138kV	Rebuild line with higher-capacity conductor	\$3,151,000	100% (\$3,151,000)
CAUSEWAY / U S ALLIANCE COOSA PINES 138kV	Rebuild line with higher-capacity conductor	\$1,088,000	100% (\$1,088,000)
U S ALLIANCE COOSA PINES / WOODVILLE 138kV	Rebuild line with higher-capacity conductor	\$2,804,000	100% (\$2,804,000)
CORNERVILLE / PUNTA DEL MAR 7302 138kV	Rebuild line with higher-capacity conductor	\$674,000	100% (\$674,000)
CORNERVILLE SUBSTATION / HALL 138kV	Rebuild line with higher-capacity conductor	\$848,000	100% (\$848,000)
HALL / TORAY PLASTIC AMERICA'S CHP PLANT 138kV	Rebuild line with higher-capacity conductor	\$424,000	100% (\$424,000)
BOYNTONVILLE / WHILLOCK 138kV	Rebuild line with higher-capacity conductor	\$1,441,000	25% (\$362,000)
	Total allocated to project		\$16,346,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.